

UNIT I - ELECTRICAL PROPERTIES OF MATERIALS

- Electrical Conductivity, Thermal conductivity – Wiedemann Franz law
- Fermi Distribution Function - Particle in a 3-Dimensional box
- Density of energy states - Classification of Solids
- Tight binding approximation
- Effective mass of an electron – Concept of Hole

1. Electrical conductivity, Thermal conductivity – Wiedemann-Franz law

Electrical Conductivity :

Definition: Coefficient of electrical conductivity (σ) which is defined as the quantity of electricity flowing per unit area per unit time maintained at unit potential gradient.
Unit: $\Omega^{-1} \text{m}^{-1}$.

Based on Drude and Lorentz theory, when we apply potential difference across the metal, the current density 'J' (which is defined as the current flowing through unit area) is given by

$$J = nev_d \quad \dots\dots\dots(1)$$

where 'n' is the free electron density and 'e' is the charge of an electron. The electrical forces acting on a free electron is given by $F = eE$

By Newton's second law of motion $F = ma$

where 'a' is the acceleration gained by the electron in the presence of the applied electric field.

Therefore,

$$ma = eE$$

$$(or) \quad a = \frac{eE}{m}$$

But acceleration = drift velocity / relaxation time = v_d / τ

$$Therefore \quad v_d = a\tau = \frac{eE\tau}{m}$$

Substituting the value of v_d in eqn.(1) we get,

$$J = \frac{neeE\tau}{m} = \frac{ne^2E\tau}{m} \quad \dots\dots \quad (2)$$

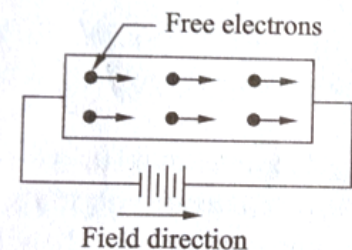
From the microscopic form of Ohm's law,

$$J = \sigma E \quad \dots\dots \quad (3)$$

Comparing the equations (2) and (3)

$$\sigma = \frac{ne^2\tau}{m} \quad \dots\dots \quad (4)$$

Thus the electrical conductivity of a metal depends on n and σ .



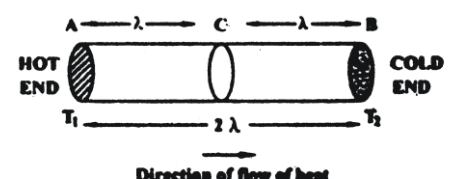
Thermal Conductivity

Definition: Thermal conductivity of the material is defined as the amount of heat conducted per unit area per unit time maintained at unit temperature gradient.

Unit: $W m^{-1}K^{-1}$

The amount of heat (Q) conducted by the rod from the end A to B of length 2λ is given by

$$Q \propto \frac{A(T_1 - T_2)t}{2\lambda}$$



(or)

$$Q = \frac{KA(T_1 - T_2)t}{2\lambda} \quad (1)$$

where $K \rightarrow$ Coefficient of thermal conductivity
 $A \rightarrow$ Area of cross section of the rod
 $(T_1 - T_2) \rightarrow$ Temperature difference between the ends A and B
 $t \rightarrow$ Time for conduction
 $2\lambda \rightarrow$ Length of rod (A to B)

From equation (1) we can write

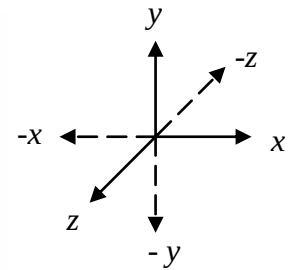
Coefficient of thermal conductivity per unit area per unit time

$$K = \frac{Q}{(T_1 - T_2)} \cdot 2\lambda \quad (2)$$

Let us assume that there is equal probability for the electrons to move in all the six directions as shown in figure. Since each electron travels with thermal velocity v , if 'n' is the free electron density, then on an average $(1/6)nv$ electrons will travel in any one direction.

Therefore, the number of electrons crossing unit area per unit time at

$$C = \frac{1}{6} nv \quad (3)$$



According to kinetic theory of gas, (Since free electrons are assumed to be gas molecules which are freely moving)

The average kinetic energy of an electron at hot end 'A' of temperature (T_1)

$$= \frac{3}{2} K_B T_1$$

The average kinetic energy of an electron at cold end 'B' of temperature (T_2)

$$= \frac{3}{2} K_B T_2$$

Where $K_B \rightarrow$ Boltzmann constant $= 1.380 \times 10^{-23} \text{ J/K}$

The heat energy transferred per unit area per unit time from end A to B across C =
 Number of electrons $\times$ Average K.E. of electron moving from A to B

$$\begin{aligned} &= \frac{1}{6} nv \cdot \frac{3}{2} K_B T_1 \\ &= \frac{1}{4} nv K_B T_1 \end{aligned} \quad (4)$$

Similarly, the heat energy transferred per unit area per unit time from end B to A across C

$$= \frac{1}{6} nv \cdot \frac{3}{2} K_B T_2$$

$$= \frac{1}{4} nv K_B T_2 \quad (5)$$

Therefore, the net heat energy transferred from end A to B per unit area per unit time across 'C' can be got by subtracting Eqn. (5) from Eqn. (4)

$$\begin{aligned} \text{i.e., } Q &= \frac{1}{4} nv K_B T_1 - \frac{1}{4} nv K_B T_2 \\ Q &= \frac{1}{4} nv K_B (T_1 - T_2) \end{aligned} \quad (6)$$

Substituting Eqn. (6) in Eqn.(2) we have
Thermal conductivity

$$\begin{aligned} K &= \frac{\frac{1}{4} nv K_B (T_1 - T_2) \cdot 2\lambda}{(T_1 - T_2)} \\ K &= \frac{nv K_B \lambda}{2} \end{aligned} \quad (7)$$

We know for metals **Relaxation time** (τ) = **Collision time** (τ_c)

$$\begin{aligned} \text{i.e., } \tau &= \tau_c = \frac{\lambda}{v} \\ \tau v &= \lambda \end{aligned} \quad (8)$$

Substituting Eqn.(8) in Eqn.(7) we have

$$K = \frac{nv K_B \tau v}{2}$$

(or) Thermal Conductivity

$$K = \frac{nv^2 K_B \tau}{2} \quad (9)$$

Eqn. (9) is the classical expression for thermal conductivity.

Wiedemann-Franz law

The ratio between the thermal conductivity and electrical conductivity of a metal is directly proportional to the absolute temperature of the metal.

$$\text{i.e., } \frac{K}{\sigma} \propto T$$

$$\text{i.e., } \frac{K}{\sigma} = LT$$

where L is a constant called as **Lorentz number**; it's value is $2.44 \times 10^{-8} \text{ W}\Omega\text{K}^{-2}$ (Quantum mechanical value) at temperature $T = 293\text{K}$.

Proof:

We know electrical conductivity (from classical theory)

$$\sigma = \frac{ne^2 \tau}{m}$$

Thermal conductivity (from classical theory)

$$K = \frac{nv^2 K_B \tau}{2}$$

Therefore

$$\frac{K}{\sigma} = \frac{\frac{nv^2 K_B \tau}{2}}{\frac{ne^2 \tau}{m}}$$

Therefore

$$\frac{K}{\sigma} = \frac{v^2 K_B m}{2e^2} \dots\dots\dots(1)$$

We know kinetic energy of an electron =

$$\frac{1}{2} mv^2 = \frac{3}{2} K_B T$$

Therefore substituting this in equation (1) we can write

$$\frac{K}{\sigma} = \frac{K_B}{e^2} \cdot \frac{3}{2} K_B T$$

$$\frac{K}{\sigma} = \frac{3}{2} \left[\frac{K_B}{e} \right]^2 T$$

$$\frac{K}{\sigma} = L T$$

Where

$$L = \frac{3}{2} \left[\frac{K_B}{e} \right]^2$$

Substituting the value of Boltzmann constant $K_B = 1.38 \times 10^{-23} \text{ JK}^{-1}$ and the charge of electron $e = 1.6021 \times 10^{-19} \text{ Joules}$, we get Lorentz Number

$$L = \frac{3}{2} \left[\frac{1.38 \times 10^{-23}}{1.6021 \times 10^{-19}} \right]^2$$

(or) $L = 1.12 \times 10^{-8} \text{ W } \Omega \text{ K}^{-2}$

It is found that the classical value of Lorentz number, is only one half of the experimental value (i.e.,) $2.44 \times 10^{-8} \text{ W } \Omega \text{ K}^{-2}$. This Discrepancy in the experimental and theoretical value of 'L' is the failure of classical theory. This discrepancy can be rectified by quantum theory.

2. Fermi Distribution Function

Definition :

Fermi function $F(E)$ represents the probability of an electron occupying a given energy state. To find out the energy states actually occupied by the free electron at any temperature (T), we can apply the Fermi-Dirac statistics. The Fermi - Dirac statistics deals with the particles (electrons) having half integral spin, named as Fermions. Thus we can write the Fermi distribution function (i.e.) the probability of an electron occupying a given energy state as

$$F(E) = \frac{1}{1 + e^{(E-E_F)/K_B T}} \quad \dots (1)$$

Where E_F is the Fermi energy, K_B is the Boltzmann constant.

Effect of temperature on fermi function

The effect of temperature on Fermi function $F(E)$ can be discussed with respect to eqn (1).

(i) At 0 Kelvin

At 0 Kelvin, the electrons can be filled only upto a maximum energy level called Fermi energy (E_{F0}), above E_{F0} all the energy levels will be empty. It can be proved from the following conditions.

i. When $E < E_F$, equation (1) becomes

$$F(E) = \frac{1}{1 + e^{-\infty}} = \frac{1}{1} = 1$$

[i.e. 100 % chance for the electron to be filled within the Fermi energy level]

ii. When $E > E_F$, equation (1) becomes

$$F(E) = \frac{1}{1 + e^{+\infty}} = \frac{1}{\infty} = 0$$

[(i.e.) 0 % chance for the electron not to be filled within the Fermi energy level]

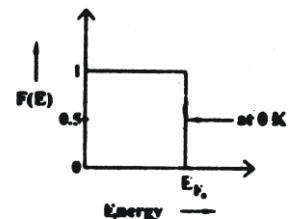
iii. When $E = E_F$, equation (1) becomes

$$F(E) = \frac{1}{1 + 1} = \frac{1}{2} = 0.5$$

[(i.e) 50 % chance for an electron to be filled and not to be filled within the Fermi energy level]

This clearly shows that at 0 Kelvin all the energy states below E_{F0} are filled and all those above it are empty.

The Fermi function at 0 kelvin (E_{F0}) can also be represented graphically as shown in figure. It is seen from the figure that the curve has step-like character at 0 kelvin.



(i.e) $F(E) = 1$ below E_{F0} and $F(E) = 0$ above E_{F0}

(ii) At any temperature T Kelvin

When the temperature is raised slowly from absolute zero, the Fermi distribution function smoothly decreases to zero as shown in figure.

Explanation:

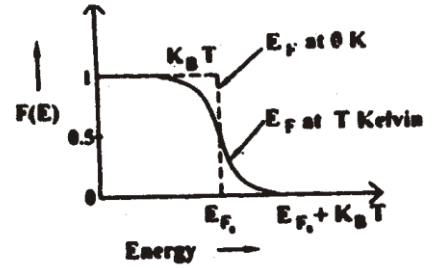
Due to the supply of thermal energy the electrons within the range of $K_B T$ below the Fermi level (E_{F0}) alone takes the energy = $K_B T$ and goes to higher energy state. Hence at any temperature (T), empty states will also be available below E_{F0} .

Electrons lying very deep (far below) the Fermi energy level (E_{F0}) will not go to excited state.

Also under classical limit

When $K_B T \ll E_F$; $F(E)$ is said to be degenerate function.

$K_B T \gg E_F$; $F(E)$ is said to be non-degenerate function.



3. Density of energy states

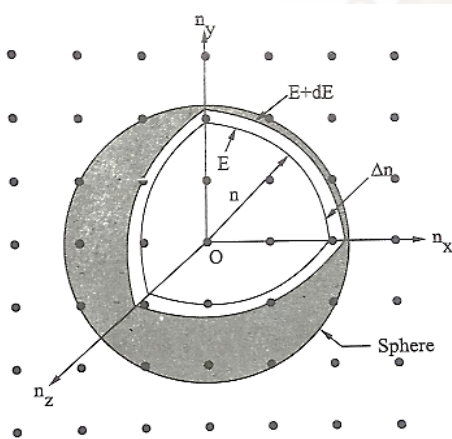
Definition:

Density of states $Z(E) dE$ is defined as the number of available electron states per unit volume in an energy interval (dE).

Explanation:

In order to fill the electrons in an energy state we have to first find the number of available energy states within a given energy interval. We know that a number of available energy levels can be obtained for various combinations of quantum numbers n_x , n_y and n_z ((i.e) $n^2 = n_x^2 + n_y^2 + n_z^2$)

Therefore, let us construct a three dimensional space of points which represents the quantum numbers n_x , n_y and n_z as shown in figure. In this space each point represents an energy level.



Number of energy levels in a cubical metal piece

To find the energy levels in a cubical metal piece and to find the number of electrons that can be filled in the given energy level, let us construct a sphere of radius 'n' in the space.

The sphere is further divided into many shells and each of this shell represents a particular combination of quantum numbers (n_x , n_y and n_z) and therefore represents a particular energy value.

Let us consider two energy values E and $E + dE$. The number of energy states between E and $E + dE$ can be found by finding the number of energy states

between the shells of radius n and $n + \Delta n$, from the origin.

The number of energy states within the sphere of radius $n = \frac{4}{3} \pi n^3$

Since n_x , n_y and n_z will have only positive values, we have to take only one octant of the sphere (i.e.) $1/8^{\text{th}}$ of the sphere volume.

$\therefore$ The number of available energy states within the sphere of radius 'n' is

$$'n' = \frac{1}{8} \left[\frac{4}{3} \pi n^3 \right]$$

Similarly, the number of available energy states within the sphere of radius ' $n + dn$ ' is

$$n + dn = \frac{1}{8} \left[\frac{4}{3} \pi (n + dn)^3 \right]$$

∴ The number of available energy states between the shells of radius n and $n + dn$ (or) between the energy levels.

$$E \text{ and } E + dE = \frac{1}{8} \left[\frac{4}{3} \pi (n + dn)^3 - \frac{4}{3} \pi n^3 \right]$$

(i.e) the number of available energy states between the energy interval dE is

$$Z(E) dE = \frac{1}{8} \frac{4\pi}{3} [n^3 + 3n^2 dn + 3n dn^2 + dn^3]$$

Since the higher powers of dn is very small, dn^2 and dn^3 terms can be neglected.

$$\therefore Z(E) dE = \frac{\pi}{6} 3n^2 dn$$

$$Z(E) dE = \frac{\pi}{2} n^2 dn \quad \dots(1)$$

We know the energy of the electron in a cubical metal piece of sides 'l'

$$E = \frac{n^2 h^2}{8ml^2}$$

$$n^2 = \frac{8ml^2 E}{h^2} \quad \dots(2)$$

$$n = \left[\frac{8ml^2 E}{h^2} \right]^{1/2} \quad \dots(3)$$

Differentiating equation (2) we get

$$2n dn = \frac{8ml^2}{h^2} dE$$

$$n dn = \frac{8ml^2}{2h^2} dE \quad \dots(4)$$

Equation (1) can be written as

$$Z(E) dE = \frac{\pi}{2} n(n dn)$$

Substituting equations (3) and (4) in the above equation we have

$$Z(E) dE = \frac{\pi}{2} \left[\frac{8ml^2 E}{h^2} \right]^{1/2} \left[\frac{8ml^2}{2h^2} dE \right]$$

$$Z(E) dE = \frac{\pi}{2} \frac{1}{2} \left[\frac{8ml^2 E}{h^2} \right]^{1/2} \left[\frac{8ml^2}{h^2} dE \right]$$

$$Z(E) dE = \frac{\pi}{2} \frac{1}{2} \left[\frac{8ml^2}{h^2} \right]^{1/2} \left[\frac{8ml^2}{h^2} \right] E^{1/2} dE$$

$$Z(E) dE = \frac{\pi}{2} \frac{1}{2} \left[\frac{8ml^2}{h^2} \right]^{3/2} E^{1/2} dE$$

$$Z(E)dE = \frac{\pi}{2} \frac{1}{2} \left[\frac{8m}{h^2} \right]^{3/2} l^3 E^{1/2} dE$$

Here l^3 represents the volume of the metal piece.

If $l^3 = 1$, then we can write that

The number of available energy states per unit volume (i.e). Density of States

$$Z(E)dE = \frac{\pi}{4} \left[\frac{8m}{h^2} \right]^{3/2} E^{1/2} dE \quad (5)$$

Since each energy level provides 2 electron states one with spin up and another with spin down (Pauli's exclusion principle), we have

$$\text{Density of states } Z(E)dE = 2 \cdot \frac{\pi}{4} \left[\frac{8m}{h^2} \right]^{3/2} E^{1/2} dE$$

$$\therefore Z(E)dE = \frac{\pi}{2} \left[\frac{8m}{h^2} \right]^{3/2} E^{1/2} dE \quad (6)$$

Carrier Concentration in metals

Let $N(E) dE$ represents the number of filled energy states between the interval of energy dE . The probability of filling of electrons in a given energy state is given by Fermi function $F(E)$.

$$N(E) dE = Z(E) dE \cdot F(E) \quad N(E)d(E) = \frac{\pi}{2} \left[\frac{8m}{h^2} \right]^{3/2} E^{1/2} dE \cdot F(E)$$

Fermi energy at 0 K

At 0 K the maximum energy level that can be occupied by the electron is known as Fermi energy level (E_0).

At 0 K for $E < E_F$; $F(E) = 1$

Integrating above equation within the limits 0 to E_{F_0} we can get the number of energy states electrons (N) within the Fermi energy E_{F_0}

$$\int_0^N N(E)d(E) = \frac{\pi}{2} \left[\frac{8m}{h^2} \right]^{3/2} \int_0^{E_{F_0}} E^{1/2} dE$$

$$N = \frac{\pi}{2} \left[\frac{8m}{h^2} \right]^{3/2} \frac{E_{F_0}^{3/2}}{3/2}$$

Number of filled energy states at 0 K

$$N = \frac{\pi}{3} \left[\frac{8m}{h^2} \right]^{3/2} E_{F_0}^{3/2}$$

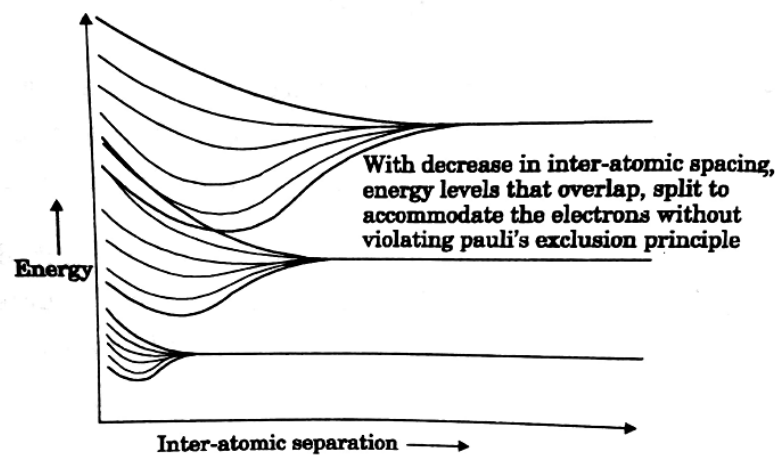
$$(\text{or}) \quad \text{Fermi Energy } E_{F_0} = \left[\frac{3N}{\pi} \right]^{2/3} \frac{h^2}{8m}$$

4. Tight binding approximation

In solid, ionic cores at fixed lattice locations and free electron gas enveloping these ionic cores. In other words, it is assumed that the solid already exists. The ionic cores are 'tightly bound' to their lattice locations. The electrons are free to run through the extend of the solid. This is called the 'Free electron approximation'.

There is another approach to modeling materials which starts from opposite position. In this approach, the atoms are independent to begin with and they are brought together to build the solid. The electrons are bound to their respective individual atoms to begin with. In this case the atoms are free to begin with while the electrons are tightly bound to the atom.

- In view of the electronic properties of the materials, this approach is referred to as the 'Tight binding approximation' - highlighting the status of the electrons at the start of the model.
- Figure shows how the tight binding approximation builds the band structure of the solid.



- When the atoms are far apart, all the bound electrons associated with each atom, have fixed energy levels.
- Assuming that building the solid starts using atoms of the same element. Thus, the energy levels occupied by the respective electrons in each atom will be identical.
- As we bring the atoms closed to each other to form the solid, the electrons will still maintain their original energy levels as long as the interatomic separation is large.
- When the atoms get close enough, the outer shell electrons begin to overlap with each other.
- The energy levels of these outer shell electrons are forced to split into energy levels above and below the energy level of these electrons when they belonged to individual atoms.
- The splitting of energy levels occurs because electrons obey the Pauli's exclusion principle.
- Initially only the outer shell electrons overlap, therefore only their levels split. But inner shell electrons still maintain their energy levels like individual atom.
- If the interatomic separation keeps decreasing even further, progressively more of the inner shell electron levels will overlap and hence also split.

- At each energy level, the level will split to enough new energy levels (band) so as to accommodate the electrons of all the atoms in the solid taken together.
- So, for example, if a hundred atoms come together, and there is one electron in the outer shell, the solid will split the energy level to a hundred energy levels. Thus the hundred outer shell electrons are filled corresponding to the combined solid.
- In view of the starting points, the free electron approximation lends itself more easily to the treatment of metallic system. The tight binding approximation is typically more consistent with the state of the material in the case of insulators, so it is better suited for modeling insulators.

5. Effective mass of electron

Effective mass of electron

The mass acquired by an electron when it is accelerated in a periodic potential is called effective mass of an electron. It is denoted by m^* .

Explanation:

When an electron is accelerated by an electrical or magnetic field in a periodic potential, the mass of the electron is not a constant. But, it varies with respect to the field applied. This varying mass is called effective mass (m^*).

Derivation of effective mass of electron

Consider a crystal subjected to an electric field of intensity 'E'. Due to this applied field, the electron gains a velocity which can be described by a wave vector k .

According to wave mechanics, a particle moving with a velocity v is equivalent to a wave packet moving with a group velocity v_g

Group velocity with which the electron can travel

$$v_g = \frac{d\omega}{dk} \quad \text{-----(1)}$$

K - wave vector, ω - angular velocity

We know that $\omega = 2\pi\nu$

$$= 2\pi \left(\frac{E}{h} \right) \text{-----(2)}$$

Substitute equation (2) in (1)

$$v_g = \frac{2\pi}{h} \left(\frac{dE}{dk} \right)$$

$$v_g = \frac{1}{\hbar} \left(\frac{dE}{dk} \right) \quad \text{-----(3)}$$

$$\text{The acceleration } a = \frac{d(v_g)}{dt}$$

$$= \frac{d}{dt} \left[\frac{1}{\hbar} \left(\frac{dE}{dk} \right) \right] \quad \text{-----(4)}$$

Multiply and divide by dk

$$a = \frac{1}{\hbar} \frac{d^2 E}{dk^2} \frac{dk}{dt}$$

Momentum $P = \frac{h}{\lambda}$ (From de-Broglie)

Multiply and divide by 2π

$$= \frac{h}{2\pi} \frac{2\pi}{\lambda}$$

$$P = \hbar k \quad \text{-----(5)} \quad \text{Since } k = \frac{2\pi}{\lambda}$$

Differentiating the equation (5) with respect to 't'

$$\frac{dp}{dt} = \hbar \frac{dk}{dt}$$

$$F = \hbar \frac{dk}{dt}$$

$$\frac{dk}{dt} = \frac{F}{\hbar} \quad \text{..... (6)}$$

Substituting equation (6) in (4)

$$a = \frac{1}{\hbar} \frac{d^2 E}{dk^2} \cdot \frac{F}{\hbar}$$

$$F = \left[\frac{\hbar^2}{\left[\frac{d^2 E}{dk^2} \right]} \right] \cdot a \quad \text{.....(7)}$$

When an electrical field is applied, acceleration of the electron due to this field

$$a = \frac{eE}{m^*} = \frac{F}{m^*} \quad [\because F = eE]$$

$$\text{(or)} \quad F = m^* a \quad \text{...(8)}$$

Comparing the equations (7) and (8), we have

$$m^* a = \left[\frac{\hbar^2}{\left(\frac{d^2 E}{dk^2} \right)} \right] a$$

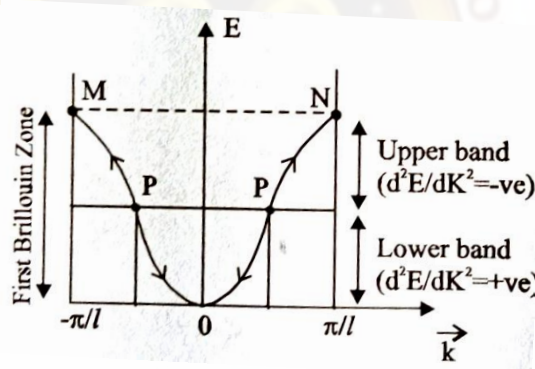
$$m^* = \frac{\hbar^2}{\left(\frac{d^2 E}{dk^2} \right)}$$

The above equation indicates that the effective mass of an electron is not a constant, but depends on the value of $\frac{d^2 E}{dk^2}$

Concept of hole (or) effective or negative mass of electron

Definition: Electron with negative effective mass is called hole (or) the electrons in the upper band behaves as a positive charged particle is called hole.

Let us take E-k curve of a single electron in a periodic potential and this E-k curve can be directed into 2 bands viz. upper and lower band w.r.to point of inflection (P).



The electron is found to exist with positive effective mass in lower band and negative effective mass (hole) in the upper band.

The advantage of hole is for a nearly filled band with 'n' number of empty states (holes) arises.

Concept of hole used to study Thomson and Hall effect.

6. Particle in a three dimensional box

The solution of one-dimensional potential well can be extended for a three-dimensional potential box. In a three-dimensional potential box, the particle can move in any direction in space. The particle can be represented by three quantum numbers n_x, n_y, n_z corresponding to x, y, z axes respectively.

Consider a particle enclosed in a three-dimensional potential box of length a, b, c along X, Y, Z axes respectively. The potential energy of the electron inside the box is constant and is taken as zero and the potential energy outside the box is infinite (∞)

The boundary conditions are

$$V(x,y,z) = 0 \quad \text{when } 0 < x < a \text{ \& } V(x,y,z) = \infty \quad 0 \geq x \geq a$$

$$V(x,y,z) = 0 \quad \text{when } 0 < y < b \text{ \& } V(x,y,z) = \infty \quad 0 \geq y \geq b$$

$$V(x,y,z) = 0 \quad \text{when } 0 < z < c \text{ \& } V(x,y,z) = \infty \quad 0 \geq z \geq c$$

The three dimensional Schrodinger time independent wave equation is

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m}{\hbar^2} [E - V] \psi = 0 \quad \dots\dots\dots(1)$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m}{\hbar^2} E \psi = 0 \quad \text{Since } V = 0 \quad \dots\dots\dots(2)$$

The solution for equation (2) can be written as

$$\psi(x,y,z) = X(x)Y(y)Z(z) \quad \dots\dots\dots(3)$$

ψ is a function of x, y and z is equal to product of three functions X, Y and Z .

The solution can be written as

$$\psi = XYZ \quad \dots\dots\dots(4)$$

Differentiating equation (4) partially with respect to 'x' twice we get

$$\frac{\partial^2 \psi}{\partial x^2} = YZ \frac{d^2 X}{dx^2} \quad \dots\dots\dots(5)$$

$$\frac{\partial^2 \psi}{\partial y^2} = XZ \frac{d^2 Y}{dy^2} \quad \dots\dots\dots(6)$$

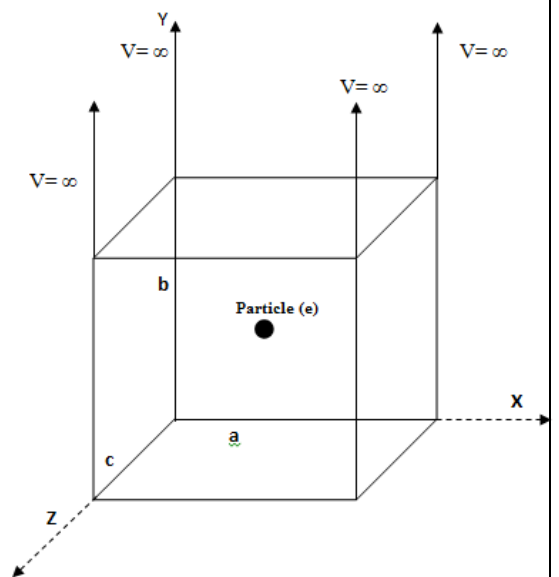
$$\frac{\partial^2 \psi}{\partial z^2} = XY \frac{d^2 Z}{dz^2} \quad \dots\dots\dots(7)$$

Substituting equations (4), (5), (6) and (7) in (2)

$$YZ \frac{d^2 X}{dx^2} + XZ \frac{d^2 Y}{dy^2} + XY \frac{d^2 Z}{dz^2} + \frac{2m}{\hbar^2} EXYZ = 0 \quad \dots\dots\dots(8)$$

Dividing by XYZ in both sides we get

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = -\frac{2mE}{\hbar^2} \quad \dots\dots\dots(9)$$



$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = -(k_x^2 + k_y^2 + k_z^2) \dots\dots\dots(10)$$

$$\frac{2mE}{\hbar^2} = (k_x^2 + k_y^2 + k_z^2) \dots\dots\dots(11)$$

In equation (10) the L.H.S is independent of each other and is equal to constants in R.H.S
By equating

$$\frac{1}{X} \frac{d^2 X}{dx^2} = -k_x^2 \quad \text{or} \quad \frac{d^2 X}{dx^2} = -k_x^2 X$$

$$\frac{d^2 X}{dx^2} + k_x^2 X = 0 \dots\dots\dots(12)$$

Similarly we get

$$\frac{d^2 Y}{dy^2} + k_y^2 Y = 0 \dots\dots\dots(13)$$

$$\frac{d^2 Z}{dz^2} + k_z^2 Z = 0 \dots\dots\dots(14)$$

Equations (12), (13) and (14) represents the differential equations in x,y,z coordinates.
The solution of (12) can be written as

$$X(x) = A_x \sin k_x x + B_x \cos k_x x \dots\dots\dots(15)$$

Where A_x and B_x are arbitrary constants

Applying the boundary conditions

(i) When $x = 0$; $X = 0$

$$\text{Eqn (15) becomes } 0 = 0 + B_x; \quad B_x = 0 \dots\dots\dots(16)$$

(ii) When $x = a$; $X = 0$

$$A_x \sin k_x a = 0; \quad B_x = 0; \quad A_x \neq 0$$

Therefore $\sin k_x a = 0 \dots\dots\dots(a)$

We know that $\sin n_x \pi = 0 \dots\dots\dots(b)$

Comparing (a) & (b) $k_x a = n_x \pi$

$$k_x = \frac{n_x \pi}{a} \dots\dots\dots(17)$$

Eqn (16) & (17) in eqn (15)

$$X(x) = A_x \sin \frac{n_x \pi x}{a} \dots\dots\dots(18)$$

Eqn (18) is unnormalized function

NORMALIZATION

Equation (18) can be normalized by integrating it within the limits (i.e) boundary condition 0 to a

$$\int_0^a |X(x)|^2 dx = 1$$

$$(or) \quad \int_0^a A_x^2 \sin^2 \frac{n_x \pi x}{a} dx = 1$$

Solving we get $\frac{A_x^2 a}{2} = 1$

$$A_x = \sqrt{\frac{2}{a}} \dots\dots\dots(19)$$

$$\text{Eqn (19) in (18)} \quad X(x) = \sqrt{\frac{2}{a}} \sin \frac{n_x \pi x}{a} \dots\dots\dots(20)$$

Similarly by solving eqn (13) & (14) with boundary conditions 0 to b and 0 to c respectively, we can write

$$Y(y) = \sqrt{\frac{2}{b}} \sin \frac{n_y \pi y}{b} \dots\dots\dots(21)$$

$$Z(z) = \sqrt{\frac{2}{c}} \sin \frac{n_z \pi z}{c} \dots\dots\dots(22)$$

Eigen Functions

The complete wave function for eqn (2) can be written as

$$\psi(x, y, z) = X(x)Y(y)Z(z) \dots\dots\dots(23)$$

Substituting eqn (20), (21) & (22) in eqn (23) will get

$$\begin{aligned} \psi_{n_x n_y n_z} &= \sqrt{\frac{2}{a}} \sin \frac{n_x \pi x}{a} \sqrt{\frac{2}{b}} \sin \frac{n_y \pi y}{b} \sqrt{\frac{2}{c}} \sin \frac{n_z \pi z}{c} \\ \psi_{n_x n_y n_z} &= \frac{2\sqrt{2}}{\sqrt{abc}} \sin \frac{n_x \pi x}{a} \sin \frac{n_y \pi y}{b} \sin \frac{n_z \pi z}{c} \dots\dots\dots(24) \end{aligned}$$

Eigen Values

From eqn (11)

$$\begin{aligned} \frac{2mE}{\hbar^2} &= (k_x^2 + k_y^2 + k_z^2) \\ E &= \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2) \dots\dots\dots(25) \end{aligned}$$

From eqn (17) we can write

$$k_x^2 = \frac{n_x^2 \pi^2}{a^2}$$

$$\text{Similarly we get } k_y^2 = \frac{n_y^2 \pi^2}{b^2}; \quad k_z^2 = \frac{n_z^2 \pi^2}{c^2}$$

Substituting these values in eqn (25) we get

$$\begin{aligned} E &= \frac{\hbar^2}{2m} \left(\frac{n_x^2 \pi^2}{a^2} + \frac{n_y^2 \pi^2}{b^2} + \frac{n_z^2 \pi^2}{c^2} \right) \\ E &= \frac{\hbar^2}{4\pi^2} \frac{\pi^2}{2m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right) \\ E &= \frac{\hbar^2}{8m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right) \dots\dots\dots(26) \end{aligned}$$

Equation (26) represents the energy Eigen value of an electron in a rectangular box.

If the box is cubical i.e the sides $a = b = c$

$$\text{We get } E_{n_x n_y n_z} = \frac{h^2}{8ma^2} (n_x^2 + n_y^2 + n_z^2) \dots\dots\dots(27)$$

And

$$\psi_{n_x n_y n_z} = \sqrt{\frac{8}{a^3}} \sin \frac{n_x \pi x}{a} \sin \frac{n_y \pi y}{a} \sin \frac{n_z \pi z}{a} \dots\dots\dots(28)$$

This shows that different combination of three quantum number (n_x, n_y, n_z) leads to different Eigen value and Eigen function.

DEGENERATE STATES:

For several combinations of quantum numbers, Eigen values are same but Eigen functions are different. Such a states are called Degenerate states.

Example:

$$n_x=1 \quad n_y=1 \quad n_z=2 \qquad n_x^2 + n_y^2 + n_z^2 = 6$$

$$n_x=1 \quad n_y=2 \quad n_z=1 \qquad n_x^2 + n_y^2 + n_z^2 = 6$$

$$n_x=2 \quad n_y=1 \quad n_z=1 \qquad n_x^2 + n_y^2 + n_z^2 = 6$$

Eigen value,

$$E_{112} = E_{121} = E_{211} = \frac{6h^2}{8ma^2}$$

Corresponding wave function

$$\psi_{112} = \sqrt{\frac{8}{a^3}} \sin \frac{\pi x}{a} \sin \frac{\pi y}{a} \sin \frac{2\pi z}{a}$$

$$\psi_{121} = \sqrt{\frac{8}{a^3}} \sin \frac{\pi x}{a} \sin \frac{2\pi y}{a} \sin \frac{\pi z}{a}$$

$$\psi_{211} = \sqrt{\frac{8}{a^3}} \sin \frac{2\pi x}{a} \sin \frac{\pi y}{a} \sin \frac{\pi z}{a}$$

NON-DEGENERATE STATES:

When only one wave function corresponds to energy Eigen value, such a states are called Non-Degenerate states.

Example:

$$\text{Eigen value } E_{222} = \frac{12h^2}{8ma^2}; \quad \text{Eigen function - } \psi_{222} = \sqrt{\frac{8}{a^3}} \sin \frac{2\pi x}{a} \sin \frac{2\pi y}{a} \sin \frac{2\pi z}{a}$$

Classification of solids:

On the basis of energy band materials are classified as Conductors, Semiconductors and Insulators.

According to band theory of solids, it consists of a set of closely spaced energy levels are called energy bands, at which electrons of different energies are going to be occupied.

Types of energy bands

- Valence band**-consists of a group of states containing outer most electrons (valence electrons).
- Conduction band**-lies above valence band with higher permitted energy bands (conduction electrons).
- Forbidden energy band** - valence band and conduction band are separated by a not allowed energy gap levels are called Forbidden energy gap.

Conductors

- Substances like copper, aluminium, silver which allow the passage of current through them are conductors.
- The valence band of these substances overlaps the conduction band as shown in fig (a).
- Due to this overlapping, a large number of free electrons are available for conduction. This is the reason, why a slight potential difference applied across them causes a heavy flow of current through them.

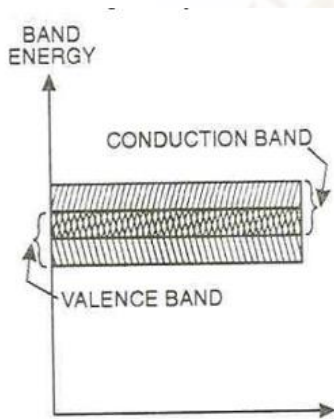


Fig. a - Conductors

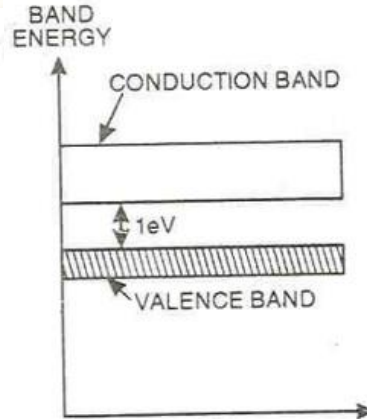


Fig. b - Semi Conductors

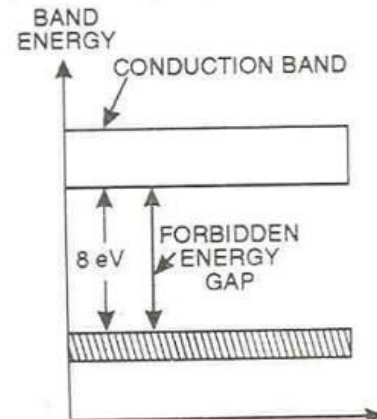


Fig. c - Insulators

Semiconductors

- Substances like carbon, silicon, germanium whose electrical conductivity lies in between the conductors and insulators are known as semiconductors.
- The valence band of these substances is almost filled, but the conduction band is almost empty.
- The forbidden energy gap between valence and conduction band is very small (1eV) as shown in fig (b).
- Therefore comparatively a smaller electric field is required to push the valence electrons to the conduction band. This is the reason, why such materials under ordinary conditions do not conduct current and behaves as an insulator.
- Even at room temperature, when some heat energy is imparted to the valence electrons, a few of them cross over to the conduction band imparting minor conductivity to the

semiconductors. As the temperature is increased, more valence electrons cross over to the conduction band and the conductivity of the material increases.

- Thus these materials have negative temperature co-efficient of resistance.

Insulators

- Substances like wood, glass, which do not allow the passage of current through them, are known as insulators.
- The valence band of these substances is full whereas the conduction band is completely empty.
- The forbidden energy gap between valence band and conduction band is very large (8eV) as shown in the fig (c).
- Therefore a large amount of energy, i.e. a very high electric field is required to push the valence electrons to the conduction band. This is the reason, why such materials under ordinary conditions do not conduct at all and are designated as insulators.

